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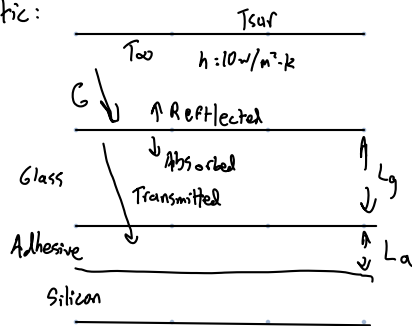
1) Known: $L_g = 3 \text{ mm}$ $h = 10 \text{ W/m}^2 \cdot \text{K}$ $T_{\text{sur}} = 300 \text{ K}$
 $2a = 0.1 \text{ mm}$ $G = 800 \text{ W/m}^2$ $h_{\text{rad}} = 5 \text{ W/m}^2 \cdot \text{K}$
 $T_{\infty} = 300 \text{ K}$ $k_g = 1.5 \text{ W/m} \cdot \text{K}$ $\eta = a - bT_{\text{si}}$
 $k_a = 1 \text{ W/m} \cdot \text{K}$ $R_c = 10^{-4} \text{ m}^2 \cdot \text{K/W}$

Find: a) Thermal Resistance Network, Value of each Resistance

b) $T_{g, \text{top}}$, T_{si}

c) η , electricity power flux

Schematic:



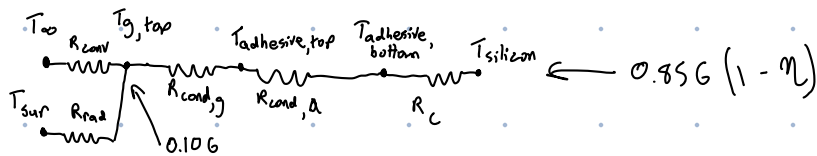
Assumptions: system in steady state

No contact resistance between glass and adhesive

Stefan Boltzman not needed for radiative losses

Analysis:

a)



$$R_{\text{conv}} = \frac{1}{h} = \frac{1}{10} = 0.1 \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

$$R_{\text{rad}} = \frac{1}{h_{\text{rad}}} = \frac{1}{5} = 0.2 \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

$$R_{\text{cond}, g} = \frac{L_g}{k_g} = \frac{0.003}{1.5} = 0.002 \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

$$R_{\text{cond}, a} = \frac{L_a}{k_a} = \frac{0.1 \text{ mm}}{1} = 1 \times 10^{-4} \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

$$R_c = 10^{-4} \frac{\text{m}^2 \cdot \text{K}}{\text{W}}$$

We assume that each node is in steady state and only the glass and silicon absorb radiation.

$$b) \quad \dot{q}_{out}'' = \frac{T_{si} - T_{\infty}}{R_{tot}} = \dot{q}_{in}'' = (0.856) \left(1 - (0.553 \cdot 0.001 T_{si}) + (0.16) \right) \\ = 680 \left(0.447 + 0.001 T_{si} \right) \\ = 303.96 + 0.68 T_{si} \text{ W}$$

$$R_{tot} = \frac{1}{\frac{1}{R_{cov}} + \frac{1}{R_{rad}}} + R_{cond,g} + R_{cond,a} + R_c =$$

$$= 0.067 + 0.002 + 1 \times 10^{-4} + 1 \times 10^{-4} = 0.0697 \frac{m^2 \cdot K}{W}$$

$$\dot{q}_{out}'' = \frac{T_{si} - 300}{0.0697} = 303.96 + 0.68 T_{si} \quad 480$$

$$T_{si} - 0.00467 T_{si} = 26.37 + 300$$

$$T_{si} = 343 \text{ K}$$

$$\dot{q}_{out}'' = 303.96 + 0.68 T_{si} + 480$$

$$= \dot{q}_{out}'' = 617.2 \text{ W}$$

$$\dot{q}_{out}'' = h (T_{g,top} - T_{\infty}) + h_{rad} (T_{g,top} - T_{\infty})$$

$$T_{g,top} = \frac{617.2}{(h + h_{rad})} + 300 \text{ K}$$

$$T_{g,top} = 341.1 \text{ K}$$

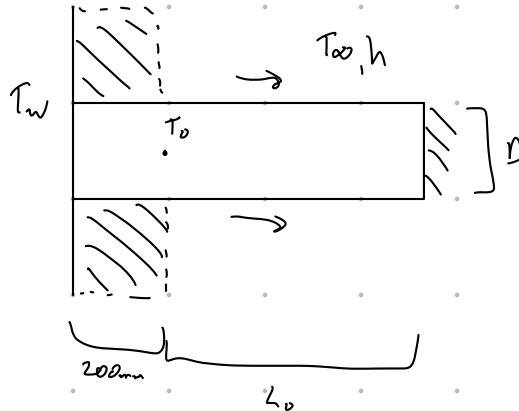
$$c) \quad \eta = a - b T_{si} = 0.553 \cdot 0.001 (343 \text{ K}) = 0.21$$

$$\dot{q}_{electriz}'' = \dot{q}_{in}'' \cdot \eta = 0.856 \cdot 0.21 = 142.8 \text{ W/m}^2$$

2) Known: $D = 10 \text{ mm}$ $T_w = 200^\circ\text{C}$ $T_{\text{max}} = 100^\circ\text{C}$ $h = 20 \text{ W/m}^2\cdot\text{K}$
 $k = 80 \text{ W/m}\cdot\text{K}$ $L_{\text{ins}} = 200 \text{ mm}$ $T_\infty = 20^\circ\text{C}$

Find: T_o , $T_o|_{L_o=200 \text{ mm}}$, L_o, min

Schematic:



Assumptions: Adiabatic tip, Rod at steady state, No radiative effects

Analysis:

$$E_{\text{in}} - E_{\text{out}} + \dot{E}_g = \dot{E}_{\text{st}}$$

$$A_c = \left(\frac{D}{2}\right)^2 \pi$$

$$P = \pi D$$

$$A_c = (L_o)(\pi D)$$

Adiabatic Tip Condition:

$$\frac{(T(x) - T_\infty)}{(T_o - T_\infty)} = \frac{\cosh[m(L_o - x)]}{\cosh(mL_o)}$$

$$Q_{\text{out}} = M \tanh(mL_o)$$

Insulated Region:

$$Q_{\text{in}} = \frac{k A_c (T_w - T_o)}{L_{\text{ins}}}$$

$$m = \sqrt{\frac{h P}{k A_c}} = \sqrt{\frac{h \pi D}{k \pi \frac{D^2}{4}}} = \sqrt{\frac{4h}{kD}}$$

$$M = \sqrt{h P k A_c} (T_o - T_\infty)$$

$$\sqrt{h k \pi D \pi \frac{D^2}{4}} (T_o - T_\infty)$$

$$\frac{1}{2} \sqrt{h k \pi^2 D^3} (T_o - T_\infty)$$

$$Q_{\text{in}} = Q_{\text{out}}:$$

$$\frac{k A_c (T_w - T_o)}{L_{\text{ins}}} = \sqrt{h P k A_c} (T_o - T_\infty) \tanh\left(\sqrt{\frac{4h}{kD}} L_o\right)$$

$$K \frac{(T_w - T_b)}{L_{ins}} = \frac{\sqrt{h P K}}{A_c} (T_b - T_{\infty}) \tanh(m L_o)$$

$$\cancel{K} \frac{(T_w - T_b)}{L_{ins}} = \cancel{K} m (T_b - T_{\infty}) \tanh(m L_o)$$

$$T_w - T_b = L_{ins} m T_b \tanh(m L_o) - m L_{ins} T_{\infty} \tanh(m L_o)$$

$$T_b (1 + m L_{ins} \tanh(m L_o)) = T_w + m L_{ins} T_{\infty} \tanh(m L_o)$$

$$T_b = \frac{T_w + m L_{ins} T_{\infty} \tanh(m L_o)}{1 + m L_{ins} \tanh(m L_o)} \quad m = \sqrt{\frac{h P}{K D}}$$

$$b) \quad L_o = 200 \text{ mm} \quad m = \sqrt{\frac{4 \cdot 20}{80 \cdot 0.01 \text{ m}}} = 10$$

$$T_b = \frac{200^\circ\text{C} + (10)(0.2 \text{ m})(20^\circ\text{C}) \tanh(10 \cdot 0.2 \text{ m})}{1 + 10(0.2 \text{ m}) \tanh(10 \cdot 0.2 \text{ m})}$$

$$T_b = 81.5^\circ\text{C} \text{ below the specified limit}$$

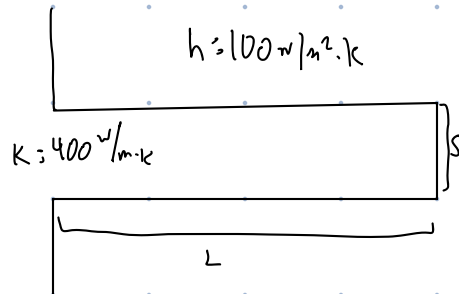
$$c) \quad T_b = 100^\circ\text{C}$$

$$L_o = 0.273 \text{ m} \quad \boxed{27.3 \text{ cm}}, \text{ from graphing function}$$

3) Known: $k = 400 \text{ W/m}\cdot\text{K}$
 $s = 10 \text{ mm}$
 $h = 100 \text{ W/m}^2\cdot\text{K}$
 $\eta_{cf} = 0.6$
 Convective tip

Find: Pin Length L , R_f , ξ_f

Schematic:



Assumptions: Convective tip, Rad. $\rightarrow$ steady state, ignore radiative effects

Analysis:

a) $m = \sqrt{\frac{hP}{kA_c}}$ $A_c = s^2 = 10^{-4} \text{ m}^2$ $A_f = P \cdot L$
 $P = 4s = 0.04$

$m = \sqrt{\frac{100 \cdot 0.04 \text{ m}}{400 \cdot 10^{-4} \text{ m}^2}} = 10$ $M = \sqrt{hPkA_c} (T_b - T_\infty)$

$\eta_{cf} = \frac{M \sinh(mL) + \left(\frac{h}{mk}\right) \cosh(mL)}{\cosh(mL) + \left(\frac{h}{mk}\right) \sinh(mL)}$

$Q_{max} = hA_f(T_b - T_\infty)$

$\eta_{cf} = \frac{mkA_c \sinh(mL) + \frac{h}{mk} \cosh(mL)}{hA_c \cosh(mL) + \frac{h}{mk} \sinh(mL)}$

$\frac{h}{mk} = \frac{100}{10 \cdot 400} = 0.025$

$\eta_{cf} = 0.6 \rightarrow L \approx 0.15 \text{ m} : \boxed{150 \text{ mm}}$, from graphing

$\eta_{cf} = \frac{Q_f}{Q_{max}}$

$$b) \quad R_f = \frac{1}{\eta_{cf} h A_c} = \frac{1}{0.6 \cdot 100 \cdot 0.04 \cdot 0.15} = 2.78 \frac{m^2}{W}$$

$$\xi_f = \frac{q_f}{q_{nf}} = \frac{\eta_f q_{max}}{q_{ft}} = \eta_f \frac{h A_c \phi_b}{h A_c \phi_b}$$

$$\xi_f = \eta_f \frac{A_f}{A_c} = \frac{0.6 \cdot 0.15 \cdot 0.004}{10^{-4}} = 36$$